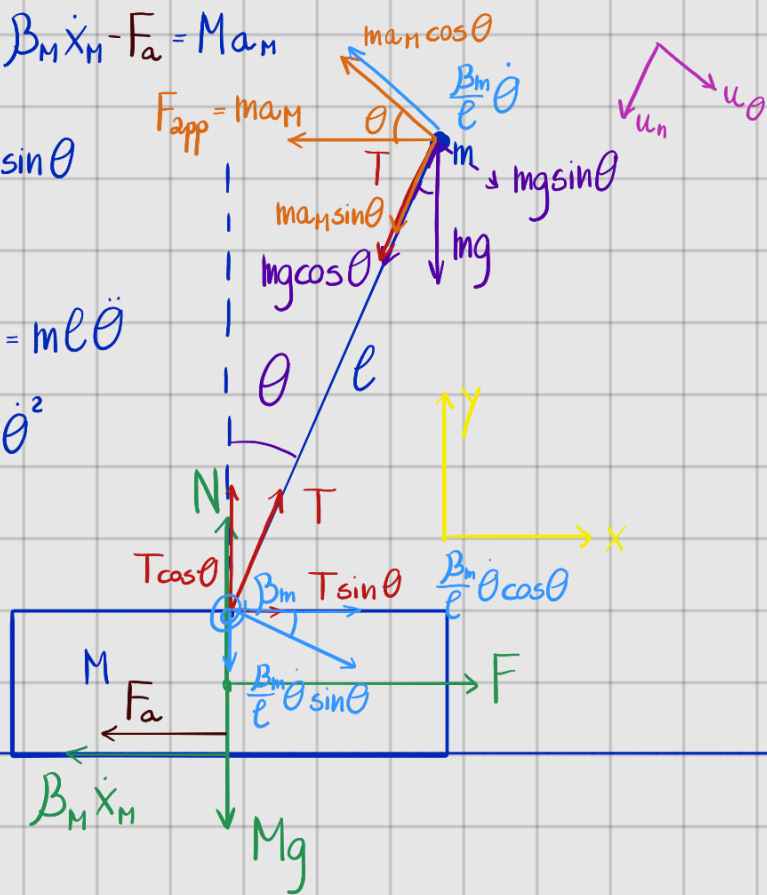


$$M: \begin{cases} x: F + \frac{B_m}{\ell} \dot{\theta} \cos \theta + T \sin \theta - B_M \dot{x}_M - F_a = M a_M \\ y: N + T \cos \theta = M g + \frac{B_m}{\ell} \dot{\theta} \sin \theta \end{cases}$$

$$m: \begin{cases} \theta: m g \sin \theta - m a_M \cos \theta - \frac{B_m}{\ell} \dot{\theta} = m \ell \ddot{\theta} \\ h: T + m g \cos \theta + m a_M \sin \theta = m \ell \dot{\theta}^2 \end{cases}$$



$$X = \begin{bmatrix} x_M \\ \dot{x}_M \\ \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$u = F$$

$$y_1 = x_M = x_1$$

$$y_2 = \theta = x_3$$

$$\begin{cases} \dot{x}_1 = \dot{x}_M = x_2 \\ \dot{x}_2 = a_M \\ \dot{x}_3 = \dot{\theta} \\ \dot{x}_4 = \ddot{\theta} \end{cases}$$

coeff. attrito
dinamico

coeff. attr.
statico

$$F_a = \left[\mu_d - (\mu_s - \mu_d) e^{-\left(\frac{|\dot{x}|}{v_s}\right)^2} \right] \cdot N \cdot \tanh(K \dot{x})$$